

Moad Lemrani

+212 6 60 98 47 99 • [✉ mouadlemrani12@gmail.com](mailto:mouadlemrani12@gmail.com)
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Profile

Second-year Computer Engineering student at ENSA Oujda, with hands-on experience building full-stack web applications and contributing to AI-powered product development. Proficient in Java, Python, and JavaScript, with a solid foundation in software architecture, computer vision, and agile workflows. Open to on-site opportunities in Morocco as well as remote positions. Seeking a summer 2026 internship to contribute to impactful engineering projects.

Education

ENSA Oujda

Engineering Degree in Computer Engineering

Relevant coursework: Data Structures & Algorithms, Object-Oriented Programming, Web Development, Machine Learning, Operating Systems, Database Systems, Software Engineering.

Oujda

Sept 2022 – Jul 2027

El Mourabitine High School

Baccalaureate in Physical Sciences – French Option

Honors: *Very Good*

Outat El Haj

Sept 2021 – Jul 2022

Technical Skills

Languages: Java, Python, PHP, JavaScript (proficient); C# (familiar)

Web & Frameworks: React, HTML5, CSS3, Symfony, FastAPI, Spring Boot, REST APIs

AI / Data: NumPy, OpenCV, Pandas, scikit-learn, Machine Learning

Tools & DevOps: Git, GitHub, Linux, Podman

Databases: MySQL, SQLite

Professional Experience

Maison d'Intelligence Artificielle (MIA)

Web Development Intern

Oujda, Morocco

Jul 2025 – Aug 2025

Project: Innomiate — Full-stack registration and team management platform for hackathon events.

- Architected the complete relational database schema and co-designed the system architecture to support concurrent multi-team workflows.
- Engineered a secure authentication pipeline (registration, CAPTCHA, email verification) using **Symfony**, eliminating unauthorized access vectors.
- Delivered 3 core functional modules: team creation, invitation system, and project pitching workflow.
- Crafted a themeable, responsive UI/UX ensuring a consistent user experience across device types.

XAI-TECH

Product Engineering Intern

Remote

Aug 2025 – Sept 2025

Project: X-Docs — Early-stage AI-powered web application enabling users to query documents through a natural-language chatbot interface.

- Defined product requirements through structured competitor analysis, identifying key differentiating features for the MVP roadmap.
- Produced comprehensive UML diagrams (use case, class, sequence) to formalize the system architecture and guide the development team.
- Built and delivered a functional no-code MVP using AI-assisted prototyping tools (V0, Lovable) to validate core user flows with stakeholders.

Academic & Personal Projects

CodeReviewer

React, Spring Boot, FastAPI, GitHub API, LLM

In Progress

- Building a full-stack AI code review tool that integrates with GitHub repositories to automatically analyze pull requests and surface actionable feedback via an LLM backend.
- Designed a microservice architecture: React frontend, Spring Boot orchestration layer, and a FastAPI service handling LLM inference and GitHub API interactions.
- Targeting automated detection of code smells, security issues, and style violations across multiple languages.

Reforestation Planning Plugin — PFA

Python, PyQGIS, QGIS

In Progress

- Developing a QGIS plugin to assist reforestation planning by analyzing geospatial data and generating site suitability maps for tree planting campaigns.
- Implementing spatial analysis algorithms in PyQGIS to process terrain, soil, and climate layers into actionable recommendations.
- Designed as a Projet de Fin d'Année (annual capstone) deliverable, combining GIS engineering with environmental impact objectives.

CBIR – Content-Based Image Retrieval

Python, NumPy, OpenCV, Tkinter

- Built a full image retrieval pipeline supporting query-by-example over large image datasets using histogram-based feature extraction.
- Benchmarked 4 similarity metrics (Histogram Intersection, Euclidean Distance, Chi-Square, Correlation) across 3 color spaces (RGB, HSV, Lab) and 3 bin configurations (8, 16, 32), achieving **97–99% retrieval precision** under optimal settings.
- Identified best-performing configuration (HSV color space, 32 bins, Chi-Square metric) through systematic empirical evaluation.

AI-Powered Money Tracker

Java, Android Studio, Groq API, Llama

- Built a native Android app for personal finance tracking with full CRUD transaction management and real-time expense visualization by category.
- Integrated an AI financial adviser powered by Meta's Llama model via the Groq API, enabling users to query their spending habits and receive personalized budget recommendations in natural language.
- Engineered local data persistence and REST API integration within a clean Android architecture.

Activities & Leadership

Path2Tech — Computer Engineering Club

ENSA Oujda

Team Leader

Oct 2024 – Jun 2025

- Directed a team of 10+ members to produce and publish 2 structured IT career roadmaps (Front-End and Back-End Web Development), collectively reaching over 200 students.

Languages

Arabic: Native

French: B2 — Upper Intermediate

English: C1 — Advanced